Guided Wave Modes: TEM, TE, TM, and Hybrid

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Lecture Objective

By the end of this lecture, you will be prepared to:

- Evaluate the modes of a waveguide using an established methodology.
- Understand the behavior of propagating and evanescent modes.

Classification of Guided Waves

- TEM modes:

$$E_z = 0$$
$$H_z = 0$$

- TE modes:

$$E_z = 0$$
$$H_z \neq 0$$

- TM modes:

$$H_z = 0$$
$$E_z \neq 0$$

- Hybrid modes:

$$E_z \neq 0$$
$$H_z \neq 0$$

Solutions to Deterministic Problem As a Superposition of Eigenmodes

- The field distribution excited within a waveguide by an arbitrary source can be represented as a superposition over a number (possibly infinite) of these modes with suitable amplitudes and phases.

Hybrid Modes

- Hybrid modes only need to be considered when the structure can support a field with both $E_z \neq 0$ and $H_z \neq 0$ that cannot be treated as a superposition of TE and TM modes.
- Hybrid modes can be supported by “hollow” tube waveguides that are partially filled with dielectric as well as dielectric waveguides like fiber optic cables.
- Detailed treatment of hybrid modes is mathematically cumbersome and beyond the scope of this class.

TEM Modes

- For TEM modes we have $E_z = H_z = 0$
- There is a non-trivial solution only when $k_c = 0$.
- The fact that $k_c = 0$ means that

$$\gamma^2 = -k^2 \quad \Rightarrow \quad \gamma = jk = j\omega\sqrt{\mu\varepsilon}$$

- From Maxwell's curl equations, we can obtain the wave impedance for TEM modes:

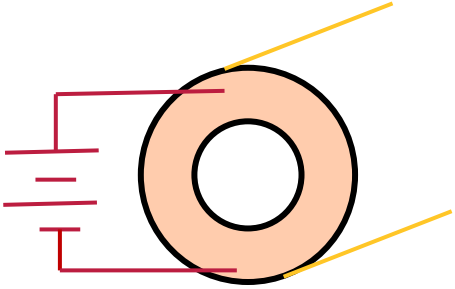
$$Z_{TEM} = \eta = \sqrt{\frac{\mu}{\varepsilon}}$$

2. TEM Modes

- We have already discussed TEM modes when we covered plane waves and transmission lines.
- Hollow tube waveguides which will be the topic of our subsequent discussions cannot support TEM modes.

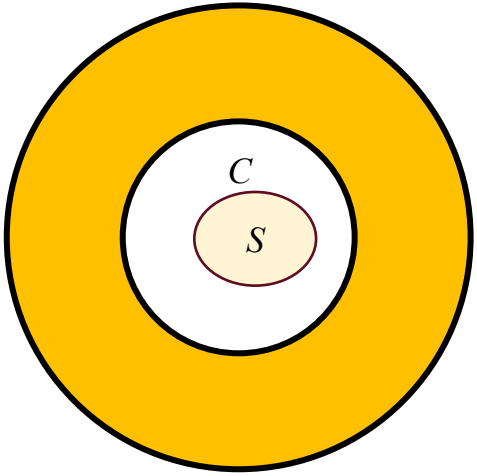
Single Conductor Waveguide Cannot Support TEM Mode

- Intuitive argument: A TEM mode must be able to propagate at zero frequency (DC).
- Consider a hollow pipe. Connect a battery to the pipe. The battery is shorted, and no signal can propagate down the pipe.



2. Single Conductor Waveguide Cannot Support TEM Mode

- Rigorous argument:
  - Assume a TEM mode exists inside the single conductor waveguide.
  - Now consider an open surface S bounded by a closed contour C in the cross-section of the waveguide.



3. Single Conductor Waveguide Cannot Support TEM Mode

- From Ampere's circuital law:

$$\oint_C \vec{H} \cdot d\,\vec{l} = \underbrace{I_{encl}}_{0} + \int_S \underbrace{\frac{\partial \vec{E}}{\partial t}}_{0} \cdot d\,\vec{s}$$

- There is no inner conductor and thus no mechanism to support conduction current within S.
- The normal to S is in the z-direction but $E_z = 0$.

- Since C is an arbitrary contour then $\vec{H}$ can have no component in the plane of the cross-section. And since $H_z = 0$, we must have $\vec{H} = 0$. Further, a time-varying electric field cannot exist without a magnetic field.
- Therefore, the TEM mode cannot exist inside a single conductor waveguide.

Question

We just described a simple experiment to demonstrate that a hollow tube waveguide cannot support TEM mode wave propagation. Can you think of simple experiment would show that a hollow tube waveguide does support some modes of electromagnetic wave propagation?

TE and TM Guided Wave Eigenmodes

- Eigenmodes are solutions to:

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{e}_z = 0 \quad + BCs$$

OR

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{h}_z = 0 \quad + BCs$$

(1)

$$\tilde{e}_x = -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial\tilde{e}_z}{\partial x} + j\omega\mu\frac{\partial\tilde{h}_z}{\partial y}\right)$$

$$\tilde{e}_y = -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial\tilde{e}_z}{\partial y} - j\omega\mu\frac{\partial\tilde{h}_z}{\partial x}\right)$$

$$\tilde{h}_x = -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial\tilde{h}_z}{\partial x} - j\omega\varepsilon\frac{\partial\tilde{e}_z}{\partial y}\right)$$

$$\tilde{h}_y = -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial\tilde{h}_z}{\partial y} + j\omega\varepsilon\frac{\partial\tilde{e}_z}{\partial x}\right)$$

(2)

Methodology for Evaluating the Modes of a Waveguide

- Determine the nature of the mode under consideration (TE or TM).
- Solve the appropriate Helmholtz equation subject to BCs to determine the longitudinal field component.
- Evaluate the transverse fields from the longitudinal field component.

TE Guided Wave Eigenmodes

- Eigenmodes are solutions to:

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{h}_z = 0 \quad + BCs$$

(1)

$$\tilde{e}_x = -\frac{1}{k_c^{\,2}}\left(j\omega\mu\frac{\partial \tilde{h}_z}{\partial y}\right)$$

$$\tilde{e}_y = -\frac{1}{k_c^{\,2}}\left(j\omega\mu\frac{\partial \tilde{h}_z}{\partial x}\right)$$

$$\tilde{h}_x = -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial \tilde{h}_z}{\partial x}\right)$$

$$\tilde{h}_y = -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial \tilde{h}_z}{\partial y}\right)$$

(2)

TM Guided Wave Eigenmodes

- Eigenmodes are solutions to:

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{e}_z = 0 \quad + BCs$$

(1)

$$\tilde{e}_x = -\frac{1}{k_c^{\,2}}\left(\gamma \frac{\partial \tilde{e}_z}{\partial x}\right)$$

$$\tilde{e}_y = -\frac{1}{k_c^{\,2}}\left(\gamma \frac{\partial \tilde{e}_z}{\partial y}\right)$$

$$\tilde{h}_x = -\frac{1}{k_c^{\,2}}\left(j\omega\varepsilon \frac{\partial \tilde{e}_z}{\partial y}\right)$$

$$\tilde{h}_y = -\frac{1}{k_c^{\,2}}\left(j\omega\varepsilon \frac{\partial \tilde{e}_z}{\partial x}\right)$$

(2)

Propagation Constant for TE and TM Waves

- Assume lossless materials:
  - Recall that $k_c = \sqrt{k^2 + \gamma^2}$ ⇒ $\gamma = \sqrt{k_c^{\,2} - k^2}$
    - The eigenvalue can be interpreted as "wavenumber at cutoff frequency":

$$k_c = \omega_c \sqrt{\mu\varepsilon} \quad \leftarrow \text{real}$$

$$k = \omega\sqrt{\mu\varepsilon} \quad \leftarrow \text{real}$$

$$\gamma = j\sqrt{k^2 - k_c^{\,2}} = jk\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}$$

$$= jk_c\sqrt{\left(\frac{k}{k_c}\right)^2 - 1} = jk_c\sqrt{\left(\frac{\omega}{\omega_c}\right)^2 - 1}$$

$$\boxed{\begin{array}{l} \omega > \omega_c \Rightarrow \gamma \text{ is purely imaginary (propagating)} \\ \omega < \omega_c \Rightarrow \gamma \text{ is purely real (evanescent)} \end{array}}$$

Cutoff Frequency

- The ***cutoff frequency*** for a given mode is f_c. We have

$$\omega_c = 2\pi f_c = \text{cutoff (radian) frequency}$$

$$k_c = \omega_c \sqrt{\mu\varepsilon} = \frac{\omega_c}{c}\sqrt{\mu_r \varepsilon_r} = \text{cutoff wavenumber}$$

2. Cutoff Frequency

- For $f > f_c$:

$$\gamma = j\beta = j\sqrt{k^2 - k_c^{\,2}}$$

$$\frac{\beta}{k_c} = \sqrt{\left(\frac{\omega}{\omega_c}\right)^2 - 1}$$

For frequencies above the cutoff frequency of the mode, the wave is propagating, i.e., its z-variation is proportional to $\exp(-j\beta z)$

3. Cutoff Frequency

- For $f < f_c$:

$$\gamma = \alpha = \sqrt{k_c^{\,2} - k^2}$$

$$\frac{\alpha}{k_c} = \sqrt{1 - \left(\frac{\omega}{\omega_c}\right)^2}$$

For frequencies below the cutoff frequency of the mode, the wave is evanescent, i.e., its z-variation is proportional to $\exp(-\alpha z)$

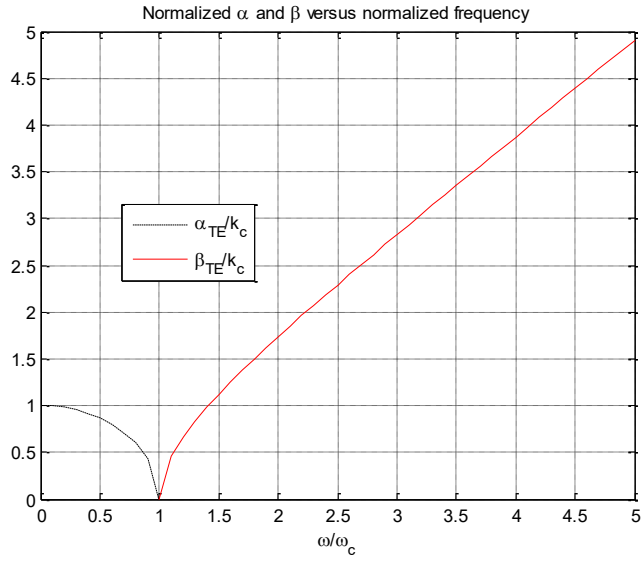
α and β versus frequency

- For $f < f_c$:

$$\frac{\alpha}{k_c} = \sqrt{1 - \left(\frac{\omega}{\omega_c}\right)^2}$$

- For $f > f_c$:

$$\frac{\beta}{k_c} = \sqrt{\left(\frac{\omega}{\omega_c}\right)^2 - 1}$$



2. α and β versus frequency

- In free space:

For $f \gg f_c$:

$$\lim_{\omega\to\infty} \frac{\beta}{k_c} = \lim_{\omega\to\infty} \sqrt{\left(\frac{\omega}{\omega_c}\right)^2 - 1} = \frac{\omega}{\omega_c}$$

$$k_c = \frac{\omega_c}{c}$$

$$\boxed{\beta \underset{\omega \gg \omega_c}{\rightarrow} \frac{\omega}{c} = k_0}$$

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